

understanding the repeated evolution of reduced plague virulence

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The recent paper by Sidhu et al. (2025) documented the repeated emergence of reduced-virulence strains of *Y. pestis*¹ across all three plague pandemics. We have been working to develop and test a mathematical model that formalizes the verbal arguments that we gave in Sidhu et al. (2025), as follows:

Our *in vivo* results showed that *pla*-reduced, attenuated strains that carry a single copy of the *pla* gene remain fully capable of causing lethal bubonic infections, inducing full blown pneumonia (as well as septicemia), contrary to engineered strains that lack *pla* or have no Pla activity (Sodeinde et al. 1992; Sebbane et al. 2006; Lathem et al. 2007; Caulfield et al. 2014; Banerjee, Crane, and Pechous 2020). However, selection for *pla* depletion likely occurred in the context of bubonic plague, as it was the only form of the disease for which we observed an effect on virulence. Knowing that *pla*-reduced strains take longer than *pla*-WT strains to kill their host in our *in vivo* experiments, it may initially appear that the selection for *pla* depletion reflects a classical transmission-virulence trade-off (Bramanti et al. 2021; Cressler et al. 2016; Lenski and May 1994; Lenski 1988). However, bubonic plague is primarily transmitted from one host to another during a very short late-stage high bacteremia, which leads to the death of the initial host and movement of their infected fleas to another host (Keeling and Gilligan 2000; Lorange et al. 2005; Hinnebusch, Jarrett, and Bland 2017). Because fatal bacteremia enhances transmission, the reduced virulence and higher host survival time due to *pla* depletion would seemingly decrease overall transmission and hence be selectively disadvantageous.

Instead, we suggest that these data support models of epidemic burnout (Parsons et al. 2024) in which pathogen attenuation enables disease persistence in small and fragmented populations through frequent, reinitiated epidemics after population recovery. Indeed, modeling has suggested that both metapopulation structure and partial population immunity could account for plague persistence in the modern

¹these strains are called “*pla*-reduced” or “*pla*-depleted”, for a reduction in copy number of the pPCP1 plasmid carrying *pla*, which encodes the plasminogen activator Pla.

plague Malagasy foci (Gascuel et al. 2013). Sylvatic rodent reservoirs are responsible for the long-term persistence of *Y. pestis* in nature (Stephen Davis et al. 2004), whereas multiple peridomestic rodent species, including black rats, play the role of amplification hosts (Zeppelini, Almeida, and Cordeiro-Estrela 2016). We focused on black rats (and their associated fleas) because they represent recognized drivers of plague epidemics in urban locales (Zeppelini, Almeida, and Cordeiro-Estrela 2016; Gage 2012; Eisen and Gage 2008; Gage and Kosoy 2005; World Health Organization 1953; Rahelinirina et al. 2010; Rahalison et al. 2003). The highly virulent *pla*-WT strain that generates frequent septicemia with a greater reproduction number than the *pla*-reduced strain would be selectively advantageous in the large, fecund naïve (Rahelinirina et al. 2010; Wu et al. 2008), highly susceptible (Eisen and Gage 2008; Gage and Kosoy 2005; World Health Organization 1953; Andrianaivoarimanana et al. 2012; Andrianaivoarimanana, Rajerison, and Jambou 2018; Tollenaere et al. 2010), and well-connected metapopulations of rats that likely characterized the early phases of pandemics and where susceptible hosts were easily replenished. However, after reductions in the size and density of peridomestic rat populations during repeated epidemic waves, we hypothesize that there was selection at the metapopulation level for *pla*-reduced strains. The high rat mortality that likely accompanied documented massive human mortality during the Black Death and the subsequent decennial epidemics likely led to low rodent densities for long periods of time. Indeed, prairie dog colonies can take years to recover after epizootics with high mortality rates of >95% (Cully and Williams 2001). Human population sizes in Europe were also reduced by ~30 to 50% during the Black Death and remained low until the mid-16th century (Bardet and Dupâquier 1997; Wrigley and Schofield 1982), likely affecting human-rat plague cycles. This decline in the density of susceptible hosts and their increasing population fragmentation about a century into the second pandemic (~1420 CE) may thus have created an environment that favored *pla*-reduced, attenuated strains of *Y. pestis* over their more virulent *pla*-WT relatives. Specifically, a decreased mortality rate and longer time to host death would allow for continued movement of rats between existing populations, now more distantly spaced than early in the pandemic, allowing for the continued persistence, and hence increased transmission, of *Y. pestis* and the subsequent slow burn of the epidemic. Although metapopulation or landscape-level mechanisms for plague persistence have been suggested (Keeling and Gilligan 2000; S. Davis et al. 2008; Schmid et al. 2012), we are not aware of previous hypotheses concerning the metapopulation-level selection of *Y. pestis*, though the concept of metapopulation-level selection for pathogen virulence or predator efficiency is well established in theoretical ecology (Claessen and Roos 1995; Boots and Sasaki 1999; Pels, Roos, and Sabelis 2002).

We were inspired by our work on epidemic burnout (Parsons et al. (2024)), which suggests that lowering within-population $\mathcal{R}_0$ (from ≈ 2) could decrease the probability of burnout in the trough after the initial introduction of plague into a population. However, the story is

more complicated.

- **demographic models matter:** Parsons et al. (2024) considered only the case where hosts are replenished at a constant rate (i.e., a constant absolute birth rate), which is reasonable for a disease with low impact on population size. For rat populations affected by plague, however, a logistic growth model (*per capita* birth rate/susceptible replenishment rate decreasing with population size; absolute birth rate highest for intermediate population size) is more reasonable. We also considered a different range of parameter values: the ratio of disease generation interval to replenishment time scale is vastly different for *Y. pestis* in rats (≈ 0.1 – 0.2) as compared to acute infectious disease (e.g. flu, measles) in humans (< 0.001).
- **simple metapopulation models may not work:** in stochastic simulation models of plague in rat metapopulations, we find a ‘ratchet’ effect over a large range of parameters. Because (1) persistence is extremely likely once an epidemic survives through the first trough and (2) a higher- $\mathcal{R}_0$, ‘wild type’ strain will outcompete a lower- $\mathcal{R}_0$, *pla*-reduced strain within a patch, the proportion of patches occupied by the wild type typically grows slowly (but irreversibly) over time. One might concoct a scenario where the accumulation is slow enough to see metapopulation-level persistence of both strains over a long time frame, but it is hard to explain takeover by a *pla*-reduced strain. Because burnout is driven by epidemic overshoot, which in turn depends on a large initial susceptible population, previous colonization of a patch by a *pla*-reduced strain actually makes it *easier* (by reducing the susceptible population size) for a wild-type strain to invade and persist in the patch.
- **seasonality may change the story:** we know that second-pandemic outbreaks in humans were seasonal (Krauer, Viljugrein, and Dean 2021). Preliminary tests suggest that adding a seasonally varying transmission rate makes it easier for a single strain to maintain intermediate patch occupancy over a long time period (rather than a ratcheted increase in occupancy as in non-seasonal models) (Figure 1). Thus such models may expand the range of parameters allowing a scenario as we originally envisioned where a *pla*-reduced strain becomes evolutionarily dominant as patch size and connectivity decrease over time.

open questions/questioning assumptions

- is the enhanced possibility of colonization (through increased survival while infected), as mentioned in Sidhu et al. (2025), sufficient to give *pla*-reduced strains an evolutionary advantage when patches are small and far apart?
- how do the changes in *pla* copy number affect rat-to-flea transmission probabilities? (not necessarily an experiment anyone wants to do ...)
- should we be considering possibilities other than the classic rat-flea transmission cycle? (evolution in humans, e.g. importance of human ectoparasite/pneumonic transmission modes?)

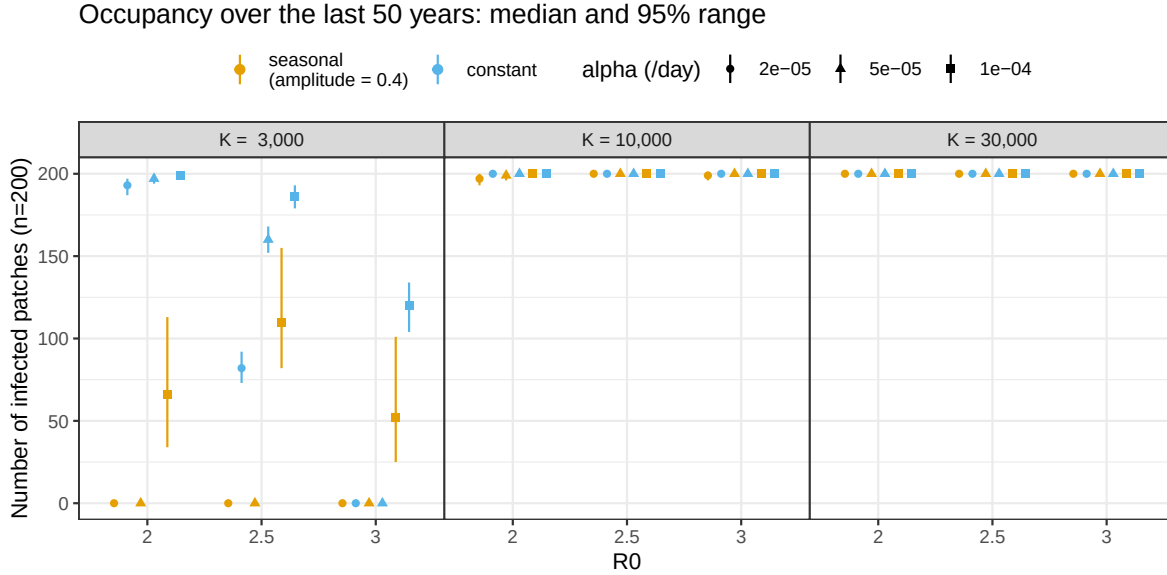


Figure 1: Long-term patch occupancy in a single-strain stochastic metapopulation model

- how can historical or ancient-DNA evidence shed light on the ancient ecology of plague and its hosts? e.g.
 - evidence of host (rat) coevolution with plague
 - finding out more about the sylvatic cycle
 - proxies for historical rat populations
 - ancient polymorphisms/substitutions that are connected to *Y. pestis* loci that have known phenotypic effects in its hosts or vectors (based on current-day empirical studies) (e.g. @ [Hinnebusch, Jarrett, and Bland 2017](#))

The general point is that ancient DNA can tell us an astonishing amount about organism-level biology, but ecology — starting with mechanisms of transmission and working up to population-level effects — is harder (although not impossible; for example, coalescent methods tell us about population history).

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